

Randomized Algorithms Final Exam

Formula Sheet

2026 Spring

Suggested use. This sheet is intended to help students recall standard probabilistic tools. It does not contain full solutions; students still need to choose the right random variables and justify each step.

1. Basic probability tools

- **Union bound:**

$$\Pr\left[\bigcup_{i=1}^t E_i\right] \leq \sum_{i=1}^t \Pr[E_i].$$

- **Independence:** for mutually independent events,

$$\Pr\left[\bigcap_{i=1}^t E_i\right] = \prod_{i=1}^t \Pr[E_i].$$

- **Conditional probability and total probability:**

$$\Pr[A \mid B] = \frac{\Pr[A \cap B]}{\Pr[B]}, \quad \Pr[A] = \sum_i \Pr[A \mid B_i] \Pr[B_i]$$

whenever $B_1, B_2, \dots$ form a partition of the sample space.

- **Useful analytic inequalities:** for $0 \leq x \leq 1$,

$$1 - x \leq e^{-x}, \quad \ln(1 - x) = -x - O(x^2), \quad H_n = \sum_{i=1}^n \frac{1}{i} = \ln n + O(1).$$

2. Expectations, indicators, and common distributions

- **Indicator variables:** if $I_E = 1$ when event E occurs and 0 otherwise, then $\mathbb{E}[I_E] = \Pr[E]$.
- **Linearity of expectation:**

$$\mathbb{E}\left[\sum_i X_i\right] = \sum_i \mathbb{E}[X_i],$$

even when the X_i 's are not independent.

- **Variance:**

$$\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

If $X_1, \dots, X_t$ are mutually independent, then

$$\text{Var}\left[\sum_{i=1}^t X_i\right] = \sum_{i=1}^t \text{Var}[X_i].$$

- **Bernoulli and binomial variables:** if $X \sim \text{Bin}(N, p)$, then

$$\Pr[X = j] = \binom{N}{j} p^j (1-p)^{N-j}, \quad \mathbb{E}[X] = Np, \quad \text{Var}[X] = Np(1-p) \leq \frac{N}{4}.$$

- **Geometric variables:** if $X \sim \text{Geom}(p)$ is the number of trials until the first success, then

$$\Pr[X = t] = (1-p)^{t-1} p, \quad \mathbb{E}[X] = \frac{1}{p}, \quad \text{Var}[X] = \frac{1-p}{p^2}.$$

3. Tail bounds

- **Markov's inequality:** for $X \geq 0$ and $a > 0$, $\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$.

- **Chebyshev's inequality:**

$$\Pr[|X - \mathbb{E}[X]| \geq a] \leq \frac{\text{Var}[X]}{a^2}.$$

- **Multiplicative Chernoff bounds:** if $X = \sum_i X_i$ is a sum of independent Bernoulli variables and $\mu = \mathbb{E}[X]$, then for $0 < \delta < 1$,

$$\Pr[X \leq (1-\delta)\mu] \leq \exp\left(-\frac{\mu\delta^2}{2}\right),$$

and for $\delta > 0$,

$$\Pr[X \geq (1+\delta)\mu] \leq \left(\frac{e^\delta}{(1+\delta)^{1+\delta}}\right)^\mu.$$

A convenient upper-tail version for $0 < \delta \leq 1$ is

$$\Pr[X \geq (1+\delta)\mu] \leq \exp\left(-\frac{\mu\delta^2}{3}\right).$$

4. Lovász Local Lemma

Symmetric Lovasz Local Lemma. Suppose bad events $A_1, A_2, \dots, A_m$ satisfy

$$\Pr[A_i] \leq p$$

and each A_i is mutually independent of all but at most d other events. If $ep(d+1) \leq 1$, then

$$\Pr\left[\bigcap_{i=1}^m \overline{A_i}\right] > 0.$$

5. Random walk / Markov-chain hitting-time equations

For a Markov chain on $\{0, 1, \dots, n\}$ with absorbing target n , let

$$h_j = \mathbb{E}[\text{time to hit } n \mid Y_0 = j].$$

The standard first-step equation is

$$h_j = 1 + \sum_k \Pr[Y_1 = k \mid Y_0 = j] h_k,$$

with boundary condition $h_n = 0$. For the pessimistic chain in the exam, $h_0 = 1 + h_1$, and for $1 \leq j \leq n-1$,

$$h_j = 1 + \frac{1}{2}h_{j-1} + \frac{1}{2}h_{j+1}.$$